



Stochastic Processes

Lecture 25

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You Cannot Beat the System!

Theorem (You Cannot Beat the System)

Let $\mathcal{G}_0 = \{\emptyset, \Omega\}$ and let $\mathcal{G} = \{\mathcal{G}_n\}_{n \in \mathbb{N}}$ be a filtration. Assume that $X_0 = 0$.

If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a **supermartingale**, and $\mathbf{H} = \{H_n\}_{n \in \mathbb{N}}$ is **non-negative and bounded predictive process** w.r.t. $\mathcal{G}$ (i.e., $|H_n| \leq L$ for all n , for some $L < +\infty$, and H_n is $\mathcal{G}_{n-1}$ -measurable for all n), then

$$\{(\mathbf{H} \cdot \mathbf{X})_n\}_{n \geq 0} \text{ is a supermartingale,}$$

with the convention that $(\mathbf{H} \cdot \mathbf{X})_0 = 0$.

Remarks: In Theorem 1, if:

- $\mathbf{X}$ is a submartingale, then $\{(\mathbf{H} \cdot \mathbf{X})_n\}_{n \geq 0}$ is a submartingale
- If $\mathbf{X}$ is a martingale, then $\{(\mathbf{H} \cdot \mathbf{X})_n\}_{n \geq 0}$ is a martingale whether or not $\mathbf{H}$ is non-negative

Stopped Supermartingale

Theorem (Stopped Supermartingale)

Let $\mathcal{G}_0 = \{\emptyset, \Omega\}$, and let $\mathcal{G} = \{\mathcal{G}_n\}_{n \geq 1}$ be a filtration.

Let $\mathbf{X} = \{X_n\}_{n \geq 0}$ be a supermartingale with respect to $\mathcal{G} = \{\mathcal{G}_n\}_{n \geq 0}$.

Let τ be a stopping time adapted to $\mathcal{G}$.

Then, $\{X_{\tau \wedge n}\}_{n \geq 0}$ is a supermartingale with respect to $\{\mathcal{G}_n\}_{n \geq 0}$.

Note: The process $\{X_{\tau \wedge n}\}_{n \geq 0}$ is called the **stopped supermartingale**.

Remarks:

- It follows from the result that

$$\mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_{\tau \wedge 0}] = \mathbb{E}[X_0] \quad \forall n.$$

Remarks on Theorem 2

- From Theorem 2, we have that

$$\mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_{\tau \wedge 0}] = \mathbb{E}[X_0] \quad \forall n.$$

- Taking limits as $n \rightarrow \infty$ on both sides, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_0].$$

Question: When can we push limits inside the expectation to conclude that $\mathbb{E}[X_\tau] \leq \mathbb{E}[X_0]$?

- If X is a martingale and τ is a stopping time, then we have from Theorem 2 that

$$\mathbb{E}[X_{\tau \wedge n}] = \mathbb{E}[X_{\tau \wedge 0}] = \mathbb{E}[X_0] \quad \forall n.$$

- Again, taking limits as $n \rightarrow \infty$ on both sides, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] = \mathbb{E}[X_0].$$

Question: When can we push limits inside the expectation to conclude that $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$?

Doob's Optional Stopping Theorem

Theorem (Doob's Optional Stopping Theorem)

Let $\mathcal{G}_0 = \{\emptyset, \Omega\}$, and let $\mathcal{G} = \{\mathcal{G}_n\}_{n \geq 1}$ be a filtration. Let $\mathbf{X} = \{X_n\}_{n \geq 0}$ be a supermartingale with respect to $\mathcal{G} = \{\mathcal{G}_n\}_{n \geq 0}$. Let τ be a stopping time adapted to $\mathcal{G}$. Suppose that one of the following conditions holds:

1. τ is **bounded a.s.**, i.e., there exists $L \in \mathbb{N}$ such that

$$\mathbb{P}(\{\tau \leq L\}) = 1.$$

2. $\mathbf{X}$ is **bounded a.s.**, i.e., there exists $K < +\infty$ such that

$$\mathbb{P}(\{|X_n| \leq K\}) = 1 \quad \forall n.$$

3. $\mathbb{E}[\tau] < +\infty$ and $\mathbf{X}$ has **bounded increments a.s.**, i.e., there exists $M < +\infty$ such that

$$\mathbb{P}(\{|X_n - X_{n-1}| \leq M\}) = 1 \quad \forall n.$$

Then, $\mathbb{E}[X_\tau] \leq \mathbb{E}[X_0]$. If $\mathbf{X}$ is a martingale and any of the conditions **1, 2, 3** holds, then $\mathbb{E}[X_\tau] = \mathbb{E}[X_0]$.

Proof of Theorem 3 (Assuming Condition 1 Holds)

- From Theorem 2, we know that $\{X_{\tau \wedge n}\}_{n \geq 0}$ is a supermartingale, and

$$\mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_0] \quad \forall n.$$

- If condition 1 holds for a sufficiently large $L \in \mathbb{N}$, setting $n = L$ above, we get

$$\mathbb{E}[X_0] \geq \mathbb{E}[X_{\tau \wedge L}] = \mathbb{E}[X_{\tau \wedge L} \mathbf{1}_{\{\tau \leq L\}}] = \mathbb{E}[X_{\tau} \mathbf{1}_{\{\tau \leq L\}}] = \mathbb{E}[X_{\tau}],$$

thus proving the desired result

- The above arguments extend similarly to the case when $\mathbf{X}$ is a martingale

Proof of Theorem 3 (Assuming Condition 2 Holds)

- From Theorem 2, we have that

$$\mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_{\tau \wedge 0}] = \mathbb{E}[X_0] \quad \forall n.$$

- Taking limits as $n \rightarrow \infty$ on both sides, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_0].$$

- Notice that $X_{\tau \wedge n} \xrightarrow{\text{a.s.}} X_\tau$
- If condition 2 holds, then applying the bounded convergence theorem, we get

$$\mathbb{E}[X_0] \geq \lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] = \mathbb{E}\left[\lim_{n \rightarrow \infty} X_{\tau \wedge n}\right] = \mathbb{E}[X_\tau].$$

- The above arguments easily extend to the case when $\mathbf{X}$ is a martingale

Proof of Theorem 3 (Assuming Condition 3 Holds)

- From Theorem 2, we have that

$$\mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_{\tau \wedge 0}] = \mathbb{E}[X_0] \quad \forall n.$$

- Taking limits as $n \rightarrow \infty$ on both sides, we get

$$\lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] \leq \mathbb{E}[X_0].$$

- Notice that $X_{\tau \wedge n} \xrightarrow{\text{a.s.}} X_\tau$
- If condition 3 holds, then

$$|X_{\tau \wedge n} - X_0| = \left| \sum_{k=1}^{\tau \wedge n} (X_k - X_{k-1}) \right| \leq M(\tau \wedge n) \leq M\tau, \quad \mathbb{E}[\tau] < +\infty.$$

- Applying the dominated convergence theorem, we get

$$\mathbb{E}[X_0] \geq \lim_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] = \mathbb{E} \left[\lim_{n \rightarrow \infty} X_{\tau \wedge n} \right] = \mathbb{E}[X_\tau].$$

- The above arguments easily extend to the case when $\mathbf{X}$ is a martingale

Miscellaneous Results

- **Corollary to Theorem 3**

If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a martingale with bounded increments, $\mathbf{H} = \{H_n\}_{n \in \mathbb{N}}$ is a bounded, predictable sequence, and τ is a stopping time with $\mathbb{E}[\tau] < +\infty$, then

$$\mathbb{E}[(\mathbf{H} \cdot \mathbf{X})_\tau] = 0.$$

- If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a nonnegative supermartingale, and τ is a stopping time, then

$$\mathbb{E}[X_\tau] \leq \mathbb{E}[X_0].$$

Proof:

— Noting that τ is finite a.s., we have

$$\mathbb{E}[X_\tau] = \mathbb{E}\left[\lim_{n \rightarrow \infty} X_{\tau \wedge n}\right] = \mathbb{E}\left[\liminf_{n \rightarrow \infty} X_{\tau \wedge n}\right] \stackrel{\text{Fatou's Lemma}}{\leq} \liminf_{n \rightarrow \infty} \mathbb{E}[X_{\tau \wedge n}] \stackrel{\text{Theorem 2}}{\leq} \mathbb{E}[X_0].$$

Convex Transformation of a Martingale

Theorem (Convex Transformation of a Martingale)

Let $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ be a martingale w.r.t. $\mathcal{G} = \{\mathcal{G}_n\}_{n \in \mathbb{N}}$, and let φ be a convex function.

Assume that $\mathbb{E}[|\varphi(X_n)|] < +\infty$ for all n . Then, the following hold.

1. The process $\{\varphi(X_n)\}_{n \in \mathbb{N}}$ is a submartingale w.r.t. $\mathcal{G}$.
2. If, additionally, φ is non-decreasing, then the result holds if $\mathbf{X}$ is a submartingale.

- The proof follows by a simple application of conditional Jensen's inequality and is left as exercise
- Consequences of Theorem 4:
 - If $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a submartingale, then $\{(X_n - a)_+\}_{n \in \mathbb{N}}$ is a submartingale for all $a \in \mathbb{R}$
 - If $\mathbf{X} = \{X_n\}_{n \in \mathbb{N}}$ is a supermartingale, then $\{X_n \wedge a\}_{n \in \mathbb{N}}$ is a supermartingale for all $a \in \mathbb{R}$

Upcrossings

- Suppose that $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a submartingale
- Fix two numbers $a < b$, and let

$$\tau^{(0)} := 0, \quad \tau^{(k)} := \begin{cases} \inf\{n \geq \tau_{k-1} : X_n \leq a\}, & k \text{ odd,} \\ \inf\{n \geq \tau_{k-1} : X_n \geq b\}, & k \text{ even.} \end{cases}$$

- $\{\tau^{(k)}\}_{k \in \mathbb{N}}$ are adapted to the natural filtration of $\mathbf{X}$, and therefore for any $m \in \mathbb{N}$,

$$\{\tau^{(2k-1)} < m \leq \tau^{(2k)}\} = \{\tau^{(2k-1)} \leq m-1\} \cap \{\tau^{(2k)} \leq m-1\}^c \in \sigma(X_0, \dots, X_{m-1}).$$

- Therefore, the process $\mathbf{H} = \{H_m\}_{m \in \mathbb{N}}$ defined by

$$H_m = \begin{cases} 1, & \exists k \in \mathbb{N} : \tau^{(2k-1)} < m \leq \tau^{(2k)}, \\ 0, & \text{otherwise,} \end{cases}$$

is a predictable process

Upcrossings

- The trajectory of $\mathbf{X}$ process from time $\tau^{(2k-1)}$ to $\tau^{(2k)}$ is called an **upcrossing**
- The net earning per upcrossing (under the previously defined gambling strategy $\mathbf{H}$) is

$$X_{\tau^{(2k)}} - X_{\tau^{(2k-1)}} \geq b - a.$$

- Let U_n denote the total number of upcrossings up to time n , i.e.,

$$U_n := \sup\{k \in \mathbb{N} : \tau^{(2k)} \leq n\}.$$

Upcrossing Lemma

Lemma (Upcrossing Lemma)

If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a submartingale, then

$$(b - a) \mathbb{E}[U_n] \leq \mathbb{E}[(X_n - a)_+] - \mathbb{E}[(X_0 - a)_+] \quad \forall n.$$

Proof:

- Make the substitution

$$Y_n = \max\{X_n, a\} = a + (X_n - a)_+, \quad n \geq 0.$$

- Whenever $\mathbf{X}$ makes an upcrossing, $\mathbf{Y}$ also makes an upcrossing
- We then have

$$\mathbb{E}[(\mathbf{H} \cdot \mathbf{Y})_n] = \mathbb{E} \left[\sum_{k=1}^n H_k (Y_k - Y_{k-1}) \right] \geq (b - a) \mathbb{E}[U_n].$$

- Letting $K_m = 1 - H_m$, we note that

$$(\mathbf{K} \cdot \mathbf{Y})_n + (\mathbf{H} \cdot \mathbf{Y})_n = Y_n - Y_0.$$

Upcrossing Lemma

Lemma (Upcrossing Lemma)

If $\mathbf{X} = \{X_n\}_{n \geq 0}$ is a submartingale, then

$$(b - a) \mathbb{E}[U_n] \leq \mathbb{E}[(X_n - a)_+] - \mathbb{E}[(X_0 - a)_+] \quad \forall n.$$

Proof:

- If $\mathbf{X}$ is a submartingale, then:
 - $\mathbf{Y}$ is a submartingale, thanks to Theorem 4
 - $\{(\mathbf{H} \cdot \mathbf{Y})_n\}_{n \geq 0}$ and $\{(\mathbf{K} \cdot \mathbf{Y})_n\}_{n \geq 0}$ are submartingales, thanks to Theorem 1
- Therefore, it follows that

$$\mathbb{E}[(\mathbf{K} \cdot \mathbf{Y})_n] \geq \mathbb{E}[(\mathbf{K} \cdot \mathbf{Y})_0] = 0,$$

and hence

$$\mathbb{E}[(\mathbf{H} \cdot \mathbf{Y})_n] = \mathbb{E}[Y_n - Y_0] - \mathbb{E}[(\mathbf{K} \cdot \mathbf{Y})_n] \leq \mathbb{E}[Y_n - Y_0].$$